The Islamia University of Bahawalpur

PROJECT PROPOSAL

Veltrixa AI

An Intelligent Digital Workspace Using Generative AI and Machine Learning

A web application for intelligent task, schedule, reminder, and productivity management

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1 Introduction

Digital work increasingly involves managing tasks, appointments, reminders, and personal goals across multiple tools. Many users still rely on disconnected notes, calendars, and to-do lists, making it difficult to turn everyday instructions into organized plans or understand their work habits.

This project proposes Veltrixa AI, an intelligent digital workspace delivered as a web application. It will allow users to manage tasks, create schedules, set reminders, and review productivity through

a unified dashboard. Machine Learning will classify user commands into supported intents, while a locally run Generative AI model will provide conversational assistance and help users interact with their workspace in natural language.

The application will focus on practical, user-controlled assistance. Users will be able to review proposed changes before the system creates or modifies tasks and schedules. The project will evaluate the command-classification component and the usefulness of the resulting web application.

2 Problem Statement

Users commonly manage daily work through separate calendars, task lists, reminder applications, and notes. This fragmentation adds friction: users must manually interpret requests, decide where information belongs, and maintain their plans across different interfaces.

A conversational interface alone does not solve this problem if it cannot reliably identify the requested action and connect it to usable task-management functions. A web-based workspace is therefore needed that combines structured task and schedule management with an intent-classification component and Generative AI assistance, while keeping users in control of changes.

3 Project Objectives

- To develop a web-based digital workspace for managing tasks, schedules, reminders, and productivity information.
- To build a Machine Learning module that classifies supported natural-language commands into workspace intents.
- To integrate a locally run Generative AI model for conversational assistance within the defined application scope.
- To convert supported user requests into reviewable task, schedule, or reminder actions.
- To provide an intelligent dashboard with clear summaries of task status and productivity activity.
- To evaluate intent classification using precision, recall, F1-score, and a confusion matrix.
- To assess usability and functional reliability through scenario-based testing and user feedback.

4 Proposed System

Veltrixa AI will provide a browser-based workspace in which a user can create and manage tasks, schedules, and reminders manually or through a conversational command interface. For example, a supported command such as “remind me to submit the report tomorrow at 10 AM” will be classified as a reminder request, its relevant details will be extracted, and the proposed reminder will be shown for confirmation before saving.

The Machine Learning component will classify commands into a defined set of intents, such as create task, set reminder, schedule activity, update task, or request a productivity summary. The

Generative AI component will support natural-language interaction and produce responses grounded in the user's workspace data and supported actions. The system will not allow the language model to make unreviewed changes to stored user records.

A productivity dashboard will present task status and activity summaries based on recorded workspace data. It will describe observed activity rather than make unsupported claims about a user's productivity or performance.

5 Main Modules

Module	Description
User Interface	Provides the workspace, task forms, calendar/schedule views, reminders, and conversational interface.
Task and Schedule Management	Creates, edits, organizes, and tracks user-approved tasks, events, and reminders.
Command Intent Classifier	Uses a trained Machine Learning model to map supported user commands to defined intents.
Command Detail Extraction	Identifies relevant details such as task text, date, time, and priority for supported commands.
Generative AI Assistant	Provides conversational help and workspace summaries using the locally run model within the project scope.
Productivity Dashboard	Displays task status, completion trends, upcoming items, and activity summaries from stored data.
Database and Authentication	Stores user accounts and workspace records using the selected database; protects user-specific data.
Evaluation and Administration	Supports test-set evaluation, error analysis, and authorized management of application settings or model artifacts.

6 Proposed Technologies

Programming Language	Python
Frontend	HTML, CSS, JavaScript
Backend	Flask
Machine Learning	Scikit-learn
Generative AI	A locally run language model and compatible local inference runtime (final choice subject to hardware availability)
Database	SQLite initially; another relational database if required
Data Handling	Pandas and NumPy
Development Environment	Visual Studio Code

Version Control	Git and GitHub

7 Proposed Methodology

The project will follow an iterative development and evaluation process:

1. Collect requirements and define the target users, supported workflows, and project boundaries.
2. Review relevant literature on digital productivity systems, intent classification, and Generative AI in user-facing applications.
3. Define the command-intent taxonomy and prepare a labeled dataset of representative user commands, including varied phrasings.
4. Clean and split the command dataset into training, validation, and held-out test subsets; prevent near-duplicate examples from leaking across splits.
5. Train and compare suitable Scikit-learn classification baselines; select the model using validation performance and error analysis.
6. Design the database schema and implement authentication, task, schedule, reminder, and dashboard functions.
7. Integrate the local Generative AI model and connect it to a limited set of application capabilities through controlled, reviewable actions.
8. Build the web interface and integrate the Machine Learning and Generative AI components with the Flask backend.
9. Test functional flows, permissions, edge cases, and model performance; collect usability feedback from test users where permitted.
10. Document results, limitations, and future work; prepare the final demonstration and deployment.

8 Expected Outcomes

The expected outcome is a working web application that unifies everyday task, schedule, reminder, and productivity-management functions with AI-assisted interaction.

- A user-friendly workspace for creating and tracking tasks, schedules, and reminders.
- A command-classification model that recognizes the intents included in the defined project scope.
- A conversational assistant powered by a locally run Generative AI model, integrated with controlled workspace functions.
- A dashboard that summarizes upcoming tasks and recorded completion activity.
- A documented evaluation of classification performance, functional reliability, and usability.

Actual classification quality and assistant usefulness will be measured during testing and will depend on the dataset, the supported command types, the selected local model, and the available hardware.

9 Project Scope

The initial version will focus on a single-user or small-scale web workspace with task management, schedules, reminders, a defined set of natural-language command intents, local Generative AI assistance, and a productivity dashboard. The project will demonstrate the feasibility of integrating these functions in one application.

The system will not attempt to replace every calendar or project-management platform, provide unrestricted autonomous computer control, or guarantee that generated responses are always correct. Integration with external calendar services, mobile applications, team collaboration, and large-scale deployment may be considered future work rather than core requirements.

10 Limitations

- Command classification performance will depend on the size, quality, and diversity of the labeled command dataset.
- Commands outside the supported intent set may be misunderstood or require manual entry.
- Generative AI responses may be inaccurate or incomplete and will require clear user review.
- Local model response quality and speed will depend on the model and the available hardware.
- Date and time interpretation may be ambiguous; the interface should request clarification or confirmation when needed.
- The productivity dashboard will reflect only the data recorded in the application and will not establish causal conclusions about productivity.
- The prototype may have limitations in scalability and integrations compared with commercial productivity platforms.

11 Significance of the Project

Veltrixa AI combines web application development, Machine Learning, and Generative AI in a practical productivity tool. It provides an opportunity to study how a command classifier and a locally run conversational model can work together in a controlled application workflow. The project also emphasizes user confirmation, measurable evaluation, and transparent limits instead of treating generated output as automatically reliable.

12 Conclusion

The proposed Veltrixa AI system aims to provide an intelligent digital workspace in which users can organize daily tasks, schedules, reminders, and productivity information through both conventional web controls and AI-assisted interaction. By combining Machine Learning-based command classification with a locally run Generative AI assistant, the project will deliver a practical web application and an evaluable academic prototype. Its success will be assessed through model

metrics, functional tests, and usability feedback, while keeping the system’s actions reviewable and its limitations explicit.